MODERN PATHOLOGY

USCAP UNITED STATES AND CANADIAN ACADEMY OF PATHOLOGY
Advancing Pathology Together

Journal homepage: https://modernpathology.org/

Research Article

Multimodal Explainable Artificial Intelligence for Prognostic Stratification of Patients With Glioblastoma

Bhakti Baheti[a,b], Sunny Rai[c,d], Shubham Innani[a], Garv Mehdiratta[e], William Robert Bell[a], Sharath Chandra Guntuku[c,d], MacLean P. Nasrallah[f], Spyridon Bakas[a,b,g,h,*]

[a] *Division of Computational Pathology, Department of Pathology and Laboratory Medicine, Indiana University School of Medicine, Indianapolis, Indiana;* [b] *Indiana University Melvin and Bren Simon Comprehensive Cancer Center, Indianapolis, Indiana;* [c] *Department of Computer and Information Science, School of Engineering and Applied Science, University of Pennsylvania, Philadelphia, Pennsylvania;* [d] *Leonard Davis Institute of Health Economics, University of Pennsylvania, Philadelphia, Pennsylvania;* [e] *School of Arts and Sciences, University of Pennsylvania, Philadelphia, Pennsylvania;* [f] *Department of Pathology and Laboratory Medicine, Perelman School of Medicine, University of Pennsylvania, Philadelphia, Pennsylvania;* [g] *Department of Neurological Surgery, Indiana University School of Medicine, Indianapolis, Indiana;* [h] *Department of Computer Science, Luddy School of Informatics, Computing, and Engineering, Indiana University, Indianapolis, Indiana*

A R T I C L E I N F O

Article history:
Received 4 October 2024
Revised 9 April 2025
Accepted 13 May 2025
Available online 24 May 2025

Keywords:
glioblastoma
histopathology
interpretability
multimodal
prognosis
XGBoost

A B S T R A C T

Glioblastoma (GBM) is the most common and aggressive malignant adult tumor of the central nervous system, with a grim prognosis and heterogeneous morphologic and molecular profiles. Since the adoption of the current standard-of-care treatment in 2005, no substantial prognostic improvement has been noticed. In this study, we seek the identification of prognostically relevant GBM characteristics from routinely acquired hematoxylin and eosin–stained whole slide images (WSIs) and clinical data, which when integrated via advanced computational methods could yield improved patient prognostic stratification and hence optimize clinical decision making and patient management. The proposed WSI analysis capitalizes on a comprehensive curation of apparent artifactual content and an interpretability mechanism via a weakly supervised attention-based multiple-instance learning approach that further utilizes clustering to constrain the search space. Patterns automatically identified by our approach as of high prognostic value classify each WSI as representative of short or long survivors. Further assessments of the prognostic relevance of the associated clinical patient data are performed both in isolation and in an integrated manner, using XGBoost and SHapley Additive exPlanations. The multimodal integration of WSI with clinical data yields enhanced stratification performance when compared with using either one of the modalities. Identifying tumor morphologic and clinical patterns associated with short and long survival will enable the clinical neuropathologist to provide additional relevant prognostic information to the treating team and suggest avenues of biological investigation for further understanding and potentially treating GBM.

Introduction

Glioblastoma (GBM) is the most common malignant adult brain tumor with heterogeneous morphologic and molecular profiles coupled with a grim prognosis (median overall survival

* Corresponding author.
E-mail address: spbakas@iu.edu (S. Bakas).

0893-3952/$ - see front matter
https://doi.org/10.1016/j.modpat.2025.100797

[OS] of 14 months).[1-3] GBM is diagnosed through the integration of histologic features seen on examination of glass slide tissue sections, molecular features, and clinical data, allowing patient treatment and management. The histologic assessment, in particular, has long stood as the reference standard for disease evaluation. Recent technologic advancements have led to the digital pathology era as conventional glass slides are more commonly scanned into high-quality digitized tissue sections, also known as whole slide images (WSIs). This transition to digitization has not only expedited the diagnostic process but also fueled the development of computational analysis techniques for histopathology imaging. As a result, new opportunities have arisen to extract valuable insights from population-based studies, promising significant advancements in our understanding of GBM.[4,5]

Prognostic stratification of patients with GBM through radiology imaging has been extensively explored.[6-10] However, these endeavors have encountered notable limitations in their ability to make precise prognostic predictions, potentially warranting a shift toward computational integrative analysis involving histopathology and clinical data. Prognostic stratification using WSI presents a formidable challenge for several reasons, including: (1) the large size of WSI (~$100,000^2$ pixels), (2) the variations within the representation of the tumor heterogeneity, and (3) availability of prognostic labels at the patient level. The literature on computational analysis of histopathology WSI can be divided into approaches analyzing the entire WSI[11] and those focusing on limited regions of interest (ROIs).[12] Given the computational complexity and resource demands associated with handling complete WSIs, ROI-based methods, particularly focusing on specific annotated ROIs identified by pathologists, have gained preference in the field.[12-15] Moreover, with the advancement of computational capabilities and increased accessibility, a multitude of supervised and unsupervised approaches continue to evolve for WSI classification.[16-18]

Numerous approaches have been developed for predicting OS from WSI for various cancer types.[15,19,20] Even modestly accurate artificial intelligence (AI) predictions of OS have demonstrated their ability to significantly enhance the statistical power of clinical trials.[21] However, many of these approaches heavily rely on handcrafted imaging features, extracted from a limited set of manually labeled WSI patches, which limits their capacity to effectively capture the diverse and heterogeneous morphology seen in GBM.[16] Some studies emphasize the quantitative performance of prognostic stratification but fall short in improving interpretability, which is essential for deepening clinical understanding of the disease.[22] Investigating and comprehending the key features that drive predictions of short and long OS can potentially enhance human expert assessments. Consequently, further focusing on the interpretation of influential features within these AI model predictions can offer additional contributions in advancing our understanding of disease and hence accelerate the path to treatment discovery.

In addition to assessing morphologic features, the diagnostic routine involves assessing patient demographic and clinical data to uncover OS-related patterns. Several approaches in the existing literature emphasize clinical data, particularly by integrating multiple data sources to predict patient OS.[23] Integrating data from various modalities, such as WSI and clinical data, can provide a more comprehensive understanding of a disease by combining information from different sources.[24,25] WSI can provide detailed representations of tissue samples revealing specific features at the microscopic level, such as cellular and nuclear atypia, proliferation, vascular features, necrosis, and infiltrative tumor cells. Clinical data, in contrast, provide high-level information about each patient, such as age, sex, medical history, and treatment outcomes. Leveraging clinical data alongside WSI enhances the accuracy of prognostic stratification and provides a deeper understanding of the factors driving disease progression and treatment response.[26,27] This can uncover new insights into disease mechanisms and potential treatment strategies. Although performance is a key factor when accounting for data from multiple modalities, the interpretability of these data is critical for revealing the underlying biology of a disease. By using clinical data in conjunction with WSI, researchers can improve the performance of prognostic stratification and gain a deeper understanding of the disease.

Although most GBMs have aggressive behavior, there is variation in patient OS. If we could better understand individual tumor behavior, the clinical team could enhance their design of personalized treatment regimens to optimize OS and quality of life for individual patients. With this motivation in mind, this study focuses on the prognostic stratification of patients with GBM between those with short and long postsurgical OS, based on patterns extracted jointly from WSI and clinical data. Our approach involves a comprehensive WSI curation to remove any known artifactual content, followed by a weakly supervised attention-based multiple-instance learning (MIL) algorithm. The interpretability mechanism of the latter, by attention heatmaps, allows morphology pattern analysis from distinct histologic GBM subcompartments of prognostic relevance. The relative and complementary benefit of integrating WSI patterns with clinical data in the prognostic stratification, through late fusion, is further assessed. Furthermore, we explored complex interactions between subgroups and patient prognosis, by examining subgroups based on *MGMT* promoter methylation status (methylated vs unmethylated), and sex (male vs female), because sex differences have shown distinct imaging and prognostic profiles.[9,28-33] Overall, this research is aimed at providing a more nuanced, interpretable, and accurate way of predicting GBM patient prognosis, potentially improving treatment decisions and patient outcomes.

Materials and Methods

Patient Population

The publicly available The cancer Genome Atlas (TCGA)-GBM[34] and TCGA-low-grade glioma (LGG)[35] data collections, available through the cancer imaging archive,[36] have been reclassified according to the latest World Health Organization (WHO) classification and used for the development and evaluation of the proposed approach. Specifically, a board-certified neuropathologist (M.P.N.), with 10 years of experience working on brain glioma, has reclassified these 2 collections according to the 2021 WHO classification of central nervous system (CNS) tumors[37] to identify all GBM isocitrate dehydrogenase (*IDH*)-wildtype (CNS WHO Gr.4) cases. The TCGA-LGG collection includes histologically low-grade astrocytomas that (based on the 2021 WHO CNS classification criteria[37]) are now classified as GBM, given the presence of molecular features that indicate the disease process and distinct tumor evolution. Thus, histologically low-grade astrocytomas from TCGA-LGG that have molecular features of GBM (previously known as "molecular GBM") are identified and included in this study's cohort to create an algorithm that applies to all clinical GBM, as currently defined by the WHO. In a related manner, specific TCGA-GBM cases have been excluded from this study's cohort as their molecular profiles are not consistent with a diagnosis of GBM, according to the current WHO criteria. The molecular features assessed for this reclassification by our neuropathologists comprised *IDH* mutation, *TERT* promoter mutation, combined gain of chromosome 7 and loss of chromosome 10, codeletion of chromosomal arms 1p and 19q, pan-glioma RNA expression cluster, *IDH*-specific RNA expression cluster, pan-glioma DNA methylation cluster, supervised DNA methylation cluster, and random forest sturm cluster. Detailed molecular information of all the cases included in our study is included in Supplementary Material 1 along with patient IDs and survival details.

After discussion with clinical experts about clinical significance and action, as well as considering equal quartiles from the median OS of the given data, the thresholds for short and long OS were set as 9 and 13 months, respectively. Cases with OS between these values were excluded to avoid unnecessary complexity during algorithmic training, while dealing with instances near the threshold boundary. As a result, this study's short- and long-survivor classes have 94 cases each, hence avoiding class imbalance issues. The rationale behind this cohort selection is illustrated step by step in Supplementary Material 2. Although our thresholds could be perceived as study specific, similar survival stratification approaches have been used in the literature, reflecting the lack of a universal survival stratification standard in GBM.[38-40] Threshold selection often depends on cohort characteristics, study design, and clinical context. The downstream experimentation is performed in a 10-fold Monte Carlo cross-validation[41] configuration, whereas the data were divided randomly and proportionally in training (80%), validation (10%), and testing (10%) data sets.

Imaging Data: Whole Slide Images

The TCGA-GBM[34] and TCGA-LGG[35] data collections comprise both formalin-fixed paraffin-embedded (FFPE) and frozen tissue section slides. As frozen sections include hydration artifacts due to freezing, this study's analysis focuses only on FFPE slides. Although multiple FFPE hematoxylin and eosin–stained WSIs are available per patient (ranging from 1 to 15), acquired from different parts of the resected tumor, we utilized only one of them to maintain uniformity across the data set. As solid tumors may have a mixture of tissue architectures and structures, resulting in an inherent heterogeneity across the available multiple WSIs, our board-certified neuropathologist (M.P.N.) selected 1 WSI representing the clinically most significant region of the tumor, following processes conventionally involved in routine clinical practice. The WSI was selected based on key criteria, including clinical relevance, tumor proportion and representation, staining quality, and the absence of artifacts, ensuring the chosen WSI was most representative of the tumor. This approach minimized variability due to sampling differences and aimed to ensure consistency in further analysis. The cases were included irrespective of their maximum apparent magnification level ($\times 20$ or $\times 40$). Of the 188 WSIs considered in this study, 107 WSIs were provided at an apparent magnification level of $\times 20$, whereas 81 WSIs were provided at $\times 40$ magnification. WSIs at $\times 40$ were then standardized by downsampling to $\times 20$ before patch extraction, whereas those at 20 were processed as is. After rescaling, patches were extracted using predefined patch sizes and grid spacing, ensuring consistent pixel resolutions across all WSIs, so that all patches represent comparable physical areas regardless of the original magnification.

Clinical Data: Molecular/Pathology Features

Clinical data of TCGA-GBM[34] and TCGA-LGG[35] comprise comprehensive patient information, including patient demographics, diagnosis details, treatment information, and molecular data collected during ongoing treatment. A total of 14 features were selected from the large pool of clinical data based on the known impact of the features in the clinical field. The selected features comprise 10 categoric features and 4 numeric features as listed in the Table 1. A categoric feature will have a fixed number of categories, and no intrinsic ordering exists between categories, eg, sex has 2 categories; (1) male and (2) female. In contrast, a numeric feature has a real-valued score, eg, Karnofsky Performance Score can have a score of 0 to 100, and the values have an ordering, ie, 100 > 90. Unfortunately, we do not have definitive information on whether the molecular studies were performed on the same tissue block as the one used for imaging analysis, but this limitation is inherent in the retrospective nature of the data set.

For subgrouping by sex analysis, there were 112 male and 76 female patients, as indicated in Table 2. Of the 188 cases, *MGMT* promoter methylation status is available for 109 patients, with 59 and 50 patients having methylated and unmethylated status, respectively. Each of this subset of data was divided into training (80%), validation (10%), and testing (10%) cohorts following 10-fold Monte Carlo cross-fold validation for experimentation.

Imaging Data Analysis

Our approach for WSI is based on a global assessment utilizing the entire WSI of each patient, rather than extracting or randomly sampling a subset of patches from a WSI (as typically done in the literature), to obtain a global/patient-level prognostic decision.[12,42] This global assessment should enable capturing the heterogeneous morphology of GBM and hence all patterns of potential prognostic relevance. The first step of the proposed approach is tissue segmentation, to distinguish the apparent tissue from the slide background. To facilitate this segmentation, the WSI is converted to the hue-saturation-value color space in a downsampled resolution (second level of the WSI pyramid, ie, downsampling by a factor of 16). A binary mask

Table 1

Features from clinical/molecular data. The clinical/molecular data comprise 10 categoric features and 4 real-valued features. The value 1 indicates a missing value

Categoric features	
1	Sex (male/female)
2	*MGMT* promoter status (methylated/unmethylated)
3	Chr 7 gain/Chr 10 loss (gain chr 7 and loss chr 10/ no combined CNA)
4	*TERT* promoter status (mutant/wildtype)
5	*BRAF* V600E status (mutant/wildtype)
6	Transcriptome subtype (CL [classical]/ NE [neural]/ ME [mesenchymal] PN [proneural])
7	Pan-glioma RNA expression cluster (LGr4/ unclassified)
8	Pan-glioma DNA Methylation cluster (LGm4/ LGm5)
9	Supervised DNA methylation cluster (classic-like/mesenchymal-like)
10	Random forest sturm cluster (mesenchymal/RTK II classic)
Numeric features	
11	Age (y at diagnosis)
12	Mutation count
13	Percent aneuploidy
14	Karnofsky performance score

CNA, copy number alteration; LG, low grade.

is created by thresholding the WSI's saturation channel, revealing the main tissue-occupied area. The mask is blurred using a median filter and then morphologically closed. This filtered tissue contour is used for further patch extraction by dividing it into rectangular patches of size 256 $\times$ 256 at apparent $\times$20 magnification level (microns per pixel range: 0.4993-0.504). The coordinates of each patch, along with its corresponding slide metadata, are stored for the feature extraction process.

Further comprehensive patch-level image curation and tissue segmentation from WSI are deemed essential to differentiate between the tissue-occupied area and artifacts. This curation should exclude patches containing artifacts, such as glass reflection, pen markings, black lines on the slide, and tissue tearing. This patch-level preprocessing involves 3 steps, as illustrated in Figure 1, with example patches demonstrating each step. First, based on the intensity values in the red-green-blue space, patches within the obtained binary mask containing substantial white background (intensity $\geq$230) or black regions (intensity $\leq$25) are discarded if the percentage of such pixels within the patch is >60%. However, this step alone cannot eliminate patches of glass reflection, dirt on the glass, or scanning artifacts. To address such scenarios, patches are converted to the hue-saturation-value space, and the percentage of pixels in "S" (intensity $\leq$25) and "V" (intensity $\geq$230) is checked. The patch is eliminated if this percentage exceeds the predefined threshold (95%). Finally, to remove patches with pen markings, the patches are converted into hematoxylin-eosin-DAB space by stain deconvolution.[43] Patches are eliminated if >80% of pixels in the eosin channel have an intensity $\leq$ 50. All selected thresholds are determined empirically, after rigorous experimentation, to ensure that only artifacts are removed and all tissue-occupied areas are preserved within the selected patches.

Figure 1 shows feature extraction for each of the retained patches, leveraging a 1024-dimensional feature vector from a pretrained ResNet-50 convolutional neural network.[44] A

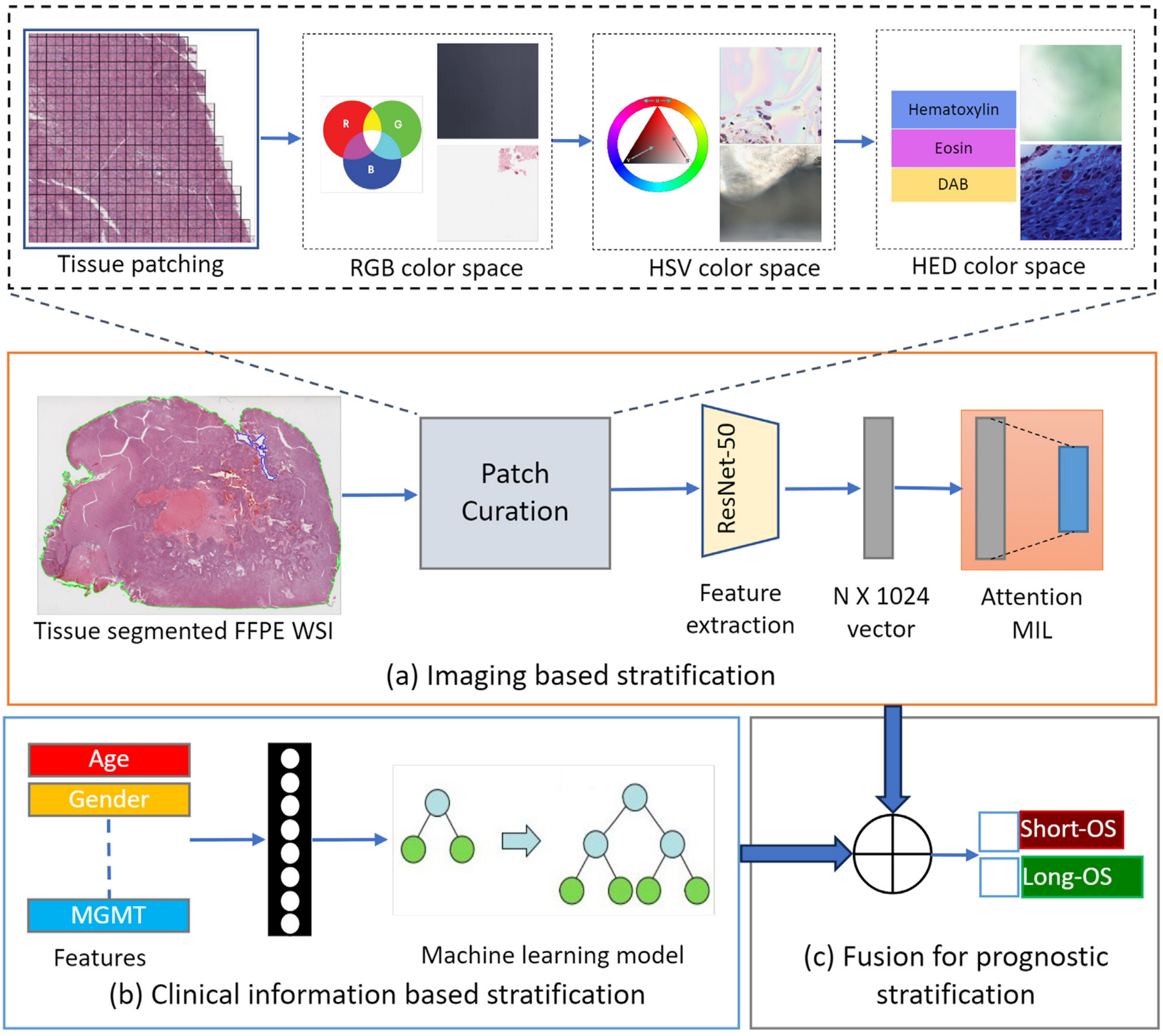

Figure 1.
Schematic representation of the complete proposed methodologic workflow. FFPE, formalin-fixed paraffin-embedded; MIL, multiple-instance learning; OS, overall survival; WSI, whole slide image.

clustering-constrained MIL framework with attention[11,45] is then used to aggregate the patch-level extracted features into slide-level representations and provide the final prediction (Fig. 1). For each class (short vs long survivor), the attention network ranks each region in the slide and assigns an "attention" score indicating its relative importance to the slide-level prediction. An additional binary clustering layer with 512 hidden neurons is introduced for learning class-specific features with support vector machine loss function after the first fully connected layer. The representative WSI patches with strong and weak attention scores train the clustering layer to learn a rich patch-level feature space, distinguishing between the positive and negative evidence of the 2 classes. The attention scores are also visualized as a heatmap to identify the important ROIs and contribute to appreciating the areas that drive the algorithmic predictions as shown in Figure 2. A dense layer is used to aggregate high-importance patch predictions in the WSI (while ignoring those of low attention score) and classify each WSI as belonging to either short or long survivors.

Clinical Data Analysis

Our analysis on clinical features is based on extreme gradient boosting (XGBoost),[46] which is an optimized distributed gradient boosting library. XGBoost builds an ensemble of decision trees sequentially, where each new tree corrects the errors of the previous ones by minimizing a loss function using gradient boosting. This ensemble tree model does not require feature scaling and can model nonlinear relationships between various features. Any missing values for any of the features from Table 1 are encoded as not-a-number. Our approach is thus robust to various data imperfections commonly found in clinical data sets, such as missing values, noise, and imbalanced classes. We trained XGBoost V 1.7.6 for prognostic stratification using the clinical and molecular features listed in Table 1. The optimal hyperparameters were identified using 10-fold stratified cross-validation on the complete data set of 188 samples. Our approach also includes regularization techniques to prevent overfitting, making it ideal for predictive modeling and risk stratification. We set the parameters, namely, "learning rate (eta)" = 0.1, "Minimum split loss (gamma)" = 0.5, "maximum depth" = 6, "subsample" = 0.6, "minimum child weight" = 2, and "L2 regularization (lambda)" = 1. These default XGBoost parameters listed here (https://xgboost.readthedocs.io/en/stable/parameter.html) provided the best performance and were fixed for the rest of the experiments.

Interpretability analysis of results obtained from predictive model is crucial for understanding the underlying factors contributing to clinical outcomes. This is enabled by XGBoost, which offers "gain" as a proxy for feature importance. Gain is practically an indication of the contribution of each feature in accuracy improvements. Additionally, we use the SHapley Additive exPlanations (SHAP)[47] approach to analyze the effect of varying feature values on the model's final predictions. SHAP is a widely used game theoretic approach to measure the importance value of each feature, representing its contribution to the model's final prediction. SHAP values are useful for explaining individual predictions as well, which is crucial in clinical settings to understand the reasoning behind a model's decision, which can impact clinical outcomes. Analyzing SHapley values with feature gain from XGBoost enables a holistic understanding of feature importance, allowing for a more transparent understanding of how complex models arrive at their predictions.

Late Fusion of Imaging and Clinical Data

Late fusion is performed to amalgamate the advantage of WSI patterns with clinical features. Note that "late fusion" in this work is following the definition given by Lipkova et al,[24] which directly refers to the aggregation of individual model predictions to generate the final model prediction. Specifically, we conduct the late fusion step by averaging the output probabilities obtained from the 2 distinct models at the decision level, using equal weights. The 2 distinct models mentioned here refer to the one model trained on the imaging features of the WSI data and the other trained on the clinical data (eg, demographic information). One of the primary advantages of late fusion is the integration of diverse data sources that leverage the strengths of each, although mitigating their individual limitations, leading to more effective and accurate outcomes.

Subgroup Analysis

With the subgrouping analysis, we intend to identify the differences in machine-generated prognoses across patient subgroups. This approach can lead to more accurate predictions by accounting for the heterogeneity across the complete patient population. One of the methods of subgrouping analysis is to group patients based on clinical features that can be used to guide treatment decisions. Here, we specifically seek prognostic stratification of patients with GBM to identify the complex interaction based on sex (male vs female) and *MGMT* promoter methylation status (methylated vs unmethylated), given the existing literature reporting on these subgroups.[28-33] Using 10 folds generated by the Monte Carlo method, we evaluate the performance of models trained using imaging and clinical data for these subgroups. Table 2 depicts the number of cases per subgroup identified for prognosis stratification.

Results

Quantitative Results of Prognostic Stratification

The quantitative performance evaluation on the validation and testing cohorts is shown in Table 3 in terms of mean stratification accuracy and the area under the curve (AUC) across 10 different folds. Specifically, Table 3 reports the average metrics of (1) multimodal model, (2) MIL model trained on only imaging data, and (3) XGBoost model trained on only clinical data for predicting prognostic stratification. It can be observed that validation of AUC values varies from 0.531 to 0.926, accompanied by accuracy scores ranging from 0.5 to 0.889. When applied to the test set, the best performance is achieved in the fifth fold with an AUC and accuracy of 0.889 underscoring its effectiveness in predicting prognostic stratification. In contrast, the worst test set performance, observed in the ninth fold, can be attributed to the challenging nature of the test cases in this specific fold, leading to lower AUC (0.494) and accuracy (0.389) scores. We observed that test cohort shows higher mean values for AUC and accuracy for the multimodal model, but the validation cohort shows higher values for model containing only imaging data. The observed discrepancy in performance between these cohorts can likely be attributed to a few factors, including the variability inherent in cross-validation methods, and the characteristics of the data in each cohort. The test and validation cohorts may differ in terms of patient

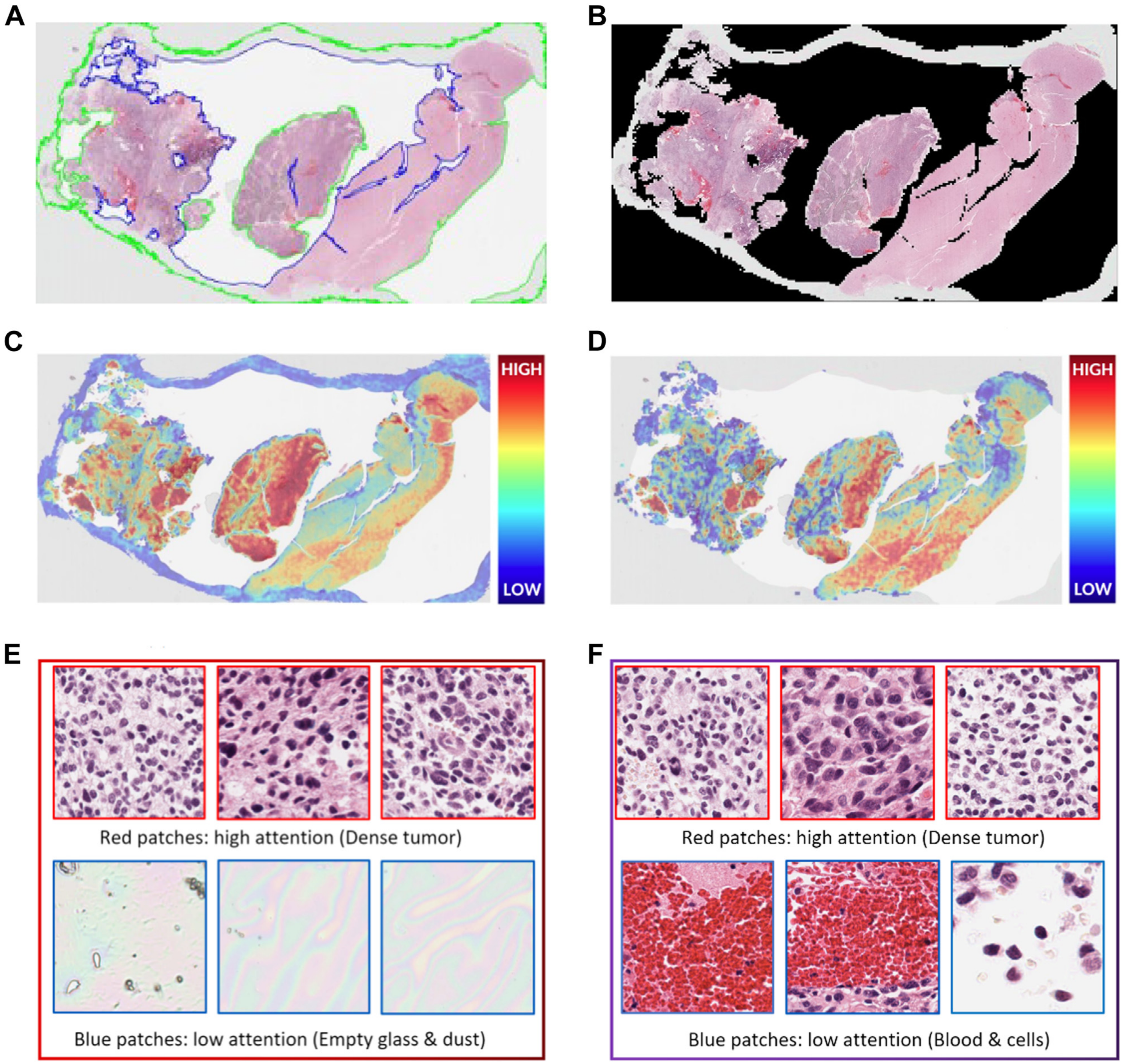

Figure 2.
Effect of preprocessing on the generated attention heatmap. (A) WSI with artifacts, (B) output of tissue segmentation, (C) heatmap without preprocessing, (D) heatmap with preprocessing, (E) sample patches without preprocessing, and (F) sample patches with preprocessing. WSI, whole slide images.

demographics, tumor characteristics, or imaging quality, which can affect the model's performance.

We also analyzed statistical significance across the 3 considered approaches (multimodal, imaging, and clinical), following the DELPHI-based recommendations for image analysis validation[48,49] incorporating (1) algorithmic ranking and (2) statistical significance testing. For this analysis, we divided the test data into multiple nonoverlapping subsets, ensuring balanced class

Table 2
Subgroup analysis results: average of 10-fold cross-validation on test cohort

Feature	Subgroup	#Samples			Imaging		Clinical		Fusion	
		Total	Short survivors	Long survivors	AUC	Accuracy	AUC	Accuracy	AUC	Accuracy
Sex	Male	112	53	59	0.620	0.527	0.617	0.6	0.633	0.627
	Female	76	41	35	0.725	0.638	0.613	0.613	0.725	0.675
MGMT	Methylated	59	35	24	0.6	0.683	0.831	0.683	0.8	0.783
	Unmethylated	50	27	23	0.717	0.660	0.667	0.640	0.750	0.7

AUC, area under the curve.

Table 3
Performance of multimodal fusion model for predicting prognostic stratification on validation and test cohorts using 10-fold Monte Carlo cross-validation method

Fold	Validation		Test	
	AUC	Accuracy	AUC	Accuracy
1	0.790	0.556	0.815	0.778
2	0.741	0.667	0.605	0.611
3	0.741	0.611	0.815	0.667
4	0.531	0.500	0.790	0.667
5	0.617	0.556	0.889	0.889
6	0.926	0.889	0.790	0.722
7	0.790	0.778	0.802	0.778
8	0.630	0.500	0.802	0.667
9	0.864	0.667	0.494	0.389
10	0.654	0.556	0.753	0.778
Mean (multimodal)	0.728	0.628	0.756	0.694
Mean (only imaging)	0.747	0.694	0.675	0.622
Mean (only clinical)	0.682	0.622	0.738	0.661

AUC, area under the curve.

Table 4
Statistical significance analysis of prognostic stratification models based on *P*-value calculation on the test data

	Multimodal	Clinical	Imaging
Multimodal	-	.1033	.0003
Clinical	-	-	.0565
Imaging	-	-	-

representation. We then computed an average rank for each of the subsets across the 3 approaches and aggregated these average rankings to produce a conclusive overall ranking. The 3 approaches were then placed in a ranked order, and their average rankings were randomly permuted (100,000 permutations), in a pairwise manner. Corresponding pairwise *P* values shown in Table 4 were computed to determine the pairwise statistical significance and report actual differences between the ordered ranked approaches. The statistical evaluation of the 3 approaches on test data revealed that multimodal approach was significantly different from only imaging approach ($P < .05$).

Qualitative Results: Attention Heatmaps and Importance of Preprocessing

Our pipeline begins with tissue segmentation, as illustrated in Figure 2A. The green boundaries delineate the regions identified as tissue by the algorithm, which are subsequently processed. In contrast, the areas enclosed by the blue boundaries represent nontissue regions such as holes that should be excluded from the analysis. Figure 2B displays the initial output of the tissue segmentation algorithm. Our approach results in the creation of high-resolution heatmaps for each WSI as shown in Figure 2C, D, effectively highlighting the crucial morphologic patterns that ultimately influence the algorithm's predictions. Figure 2E, F shows some sample patches where the algorithm has given high attention and low attention while making prediction.

Subgrouping Model Results

We report the average performance metrics of the model for sex—male, female; *MGMT*—methylated, unmethylated, for imaging, clinical, and fusion modalities in Table 2, which offers a comprehensive overview of subgroup results derived from a 10-fold cross-validation analysis, and demonstrates the superiority of the fusion models over individual models. In the context of sex-based subgroups, the results reveal distinctive performance patterns. As it can be observed from Table 2, in male participants, the fusion model outperforms the imaging and clinical models with an AUC of 0.633 and an accuracy of 0.627. For female participants, the fusion model also shows superiority, with an AUC of 0.725 and an accuracy of 0.675, surpassing the other models. The performance of different models was also evaluated based on *MGMT* methylation status subgroups. In the methylated subgroup, the fusion model performed best in terms of accuracy (0.783), whereas the clinical model also showed strong results with AUC of 0.831. The unmethylated subgroup exhibited different trends, with the fusion model again excelling (AUC = 0.75), but the imaging model showed better performance (AUC = 0.717) than clinical model (AUC = 0.667).

This detailed subgroup analysis provides valuable insights into how different models perform across subgroups, shedding light on the role of specific attributes, such as sex and *MGMT* methylation status, in influencing model performance.

Discussion

In this study, we introduced a novel approach for prognostic stratification of patients with GBM, leveraging the power of computational algorithms toward integrated diagnostics. Prognostic stratification through the multimodal analysis of WSI and clinical data represents an active and promising area within the domain of the health care AI field. The fusion of these 2 distinct data models proves to enhance the stratification accuracy compared with individual models and enables a comprehensive understanding of a patient's disease and prognosis. WSI analysis offers valuable insights into the microscopic morphologic characteristics of the underlying tissue. However, analysis of clinical data, encompassing demographic and other relevant information, helps to identify high-level heuristic patient characteristics. Beyond the stratification accuracy, this study also focuses on the interpretability of the obtained predictions to understand what drives particular algorithmic decisions and attempts to further contribute to our clinical understanding of this disease. Interpretations from the results are discussed in the following sections.

Imaging Analysis: Interpretation of Attention Heatmaps

The core of our imaging approach relies on a clustering-constrained attention-based MIL algorithm, which effectively categorizes WSIs into short and long survivors and also assigns attention scores to individual patches within each WSI to generate high-resolution heatmap. The image curation steps as described in Materials and Methods to eliminate various artifacts, such as glass reflections, dust, and pen markings, play a vital role in creating the focused heatmaps to improve the interpretability of results. It is evident from the tissue segmentation output in Figure 2B that the algorithm initially erroneously designated some artifactual background content as tissue; the corresponding attention heatmap in Figure 2C demonstrates that low attention is given to the misdesignated artifactual region. Given that the low attention areas in

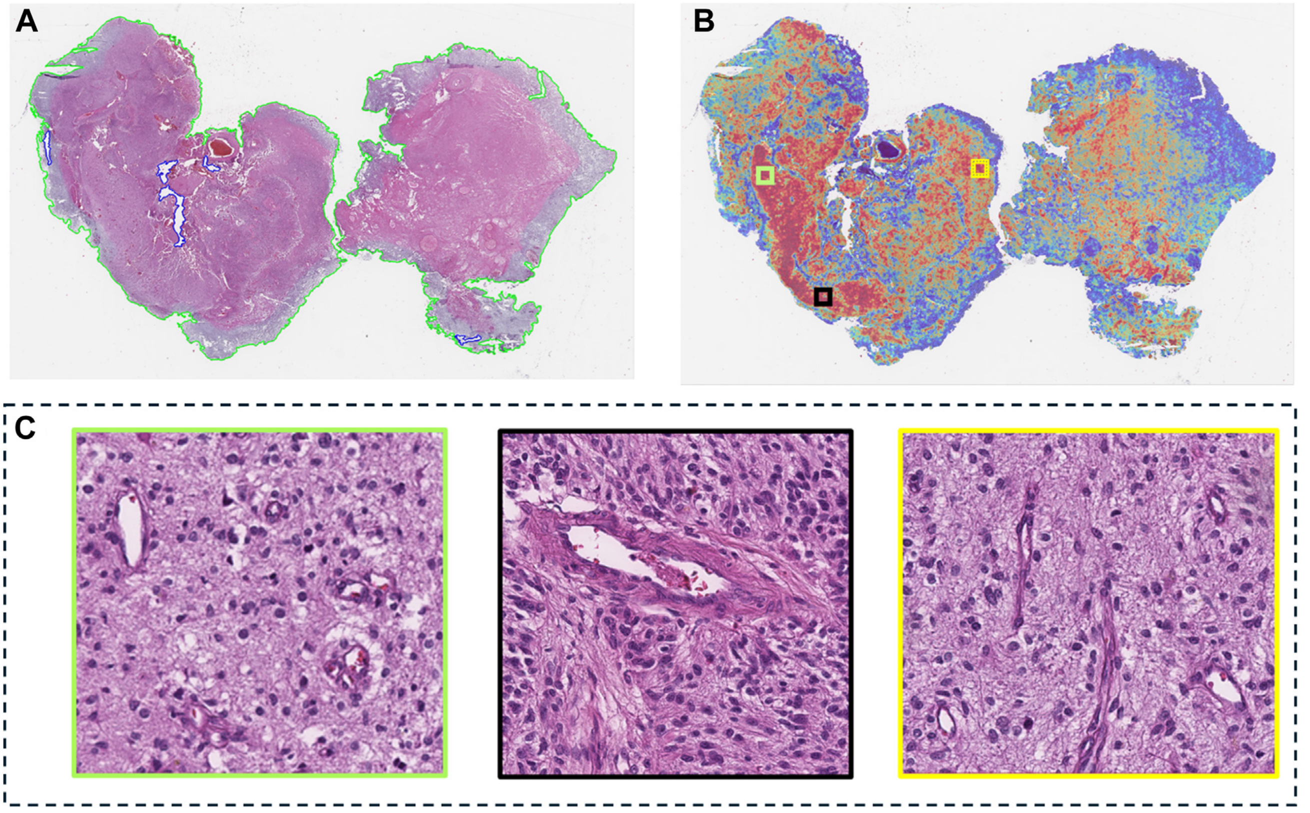

Figure 3.
Interpretation of the high-attention region from WSI of a correctly predicted short survivor. The model has given high attention to infiltrative tumor with lower cellular density and microvascular proliferation indicative of aggressive tumor. (A) Original tissue-segmented WSI, (B) attention heatmap, and (C) sample patches from high-attention region of the heatmap. Color of the boundary indicates respective region from the heatmap. WSI, whole slide images.

Figure 2E are actually “nontissue” regions, they do not convey meaningful information about the underlying morphology. Ideally, this region should have been entirely excluded from the analysis. In some other cases, the interpretability algorithm gives significant attention to pen markings, which is an undesirable influence on the decision. However, the new heatmap in Figure 2D with incorporated curation steps is more accurate and focused highlighting the patches of low attention (Fig. 2F) in a more meaningful manner. Thus, the proposed preprocessing steps enhance the overall reliability of our analysis.

The attention heatmaps effectively highlight the crucial morphologic patterns that ultimately influence the algorithm’s predictions. By interpreting these heatmaps derived from short and long survivors, we can pinpoint the critical histologic regions within the WSIs that hold significant prognostic value. This visualization of heatmap offers valuable insights into the interpretability of the algorithmic decisions, enhancing our understanding of histologic regions with prognostic significance. This comprehensive approach allows us to capture the heterogeneous morphology of GBM effectively and ensures that we consider all potential patterns of prognostic relevance. Figure 3 shows one of the WSI from the testing cohort with a corresponding heatmap that aids in visualizing and interpreting underlying morphologic patterns and their relative importance. This WSI is from a short survivor and is correctly predicted as a short survivor by our algorithm. Figure 3C also shows some of the sample patches with high attention, ie, identified as of high importance for the final decision by the algorithm itself. The morphologic patterns identified as of high diagnostic importance from the heatmap are discussed with expert neuropathologists (M.P.N. and W.R.B.) to understand if they also recognize them as clinically significant. The patterns were indeed indicative of aggressive tumor showing increased tumor infiltration and microvascular proliferation. In addition, we discussed correctly predicted long-survivor cases as well as misclassified short- and long-survivor cases for precise interpretation of the algorithmic decisions. From such heatmap interpretations with neuropathologists, we discovered that for correctly predicted long survivors, the heatmaps indicate high attention throughout histologically malignant areas, including necrosis, hypercellularity, atypia, infiltration, and proliferation. For an incorrectly predicted long survivor, necrotic area was not heavily weighted, and we observed densely packed gemistocytic cells. For another correctly predicted short survivor, infiltration of gemistocytic cells without clear malignant features was present. Although the leptomeninges was generally not highly weighted on the heatmaps, foci within them containing the concerned atypical cells potentially indicative of the leptomeningeal spread of tumor were highly weighted. After this qualitative investigation, we found that the attention heatmaps can often be reconciled with the pathologist’s opinion, but indepth analysis is necessary for a deeper understanding. In addition, these heatmaps can also be used to rigorously analyze the misclassified cases and analyze their characteristics. Although we included the qualitative assessment of attention heatmaps in the main article, the quantitative data-driven analysis with unsupervised uniform manifold approximation and projection[50] clustering is included in Supplementary Material 3, which provides insights into consistent high-attention regions for short- and long-term survivors. The

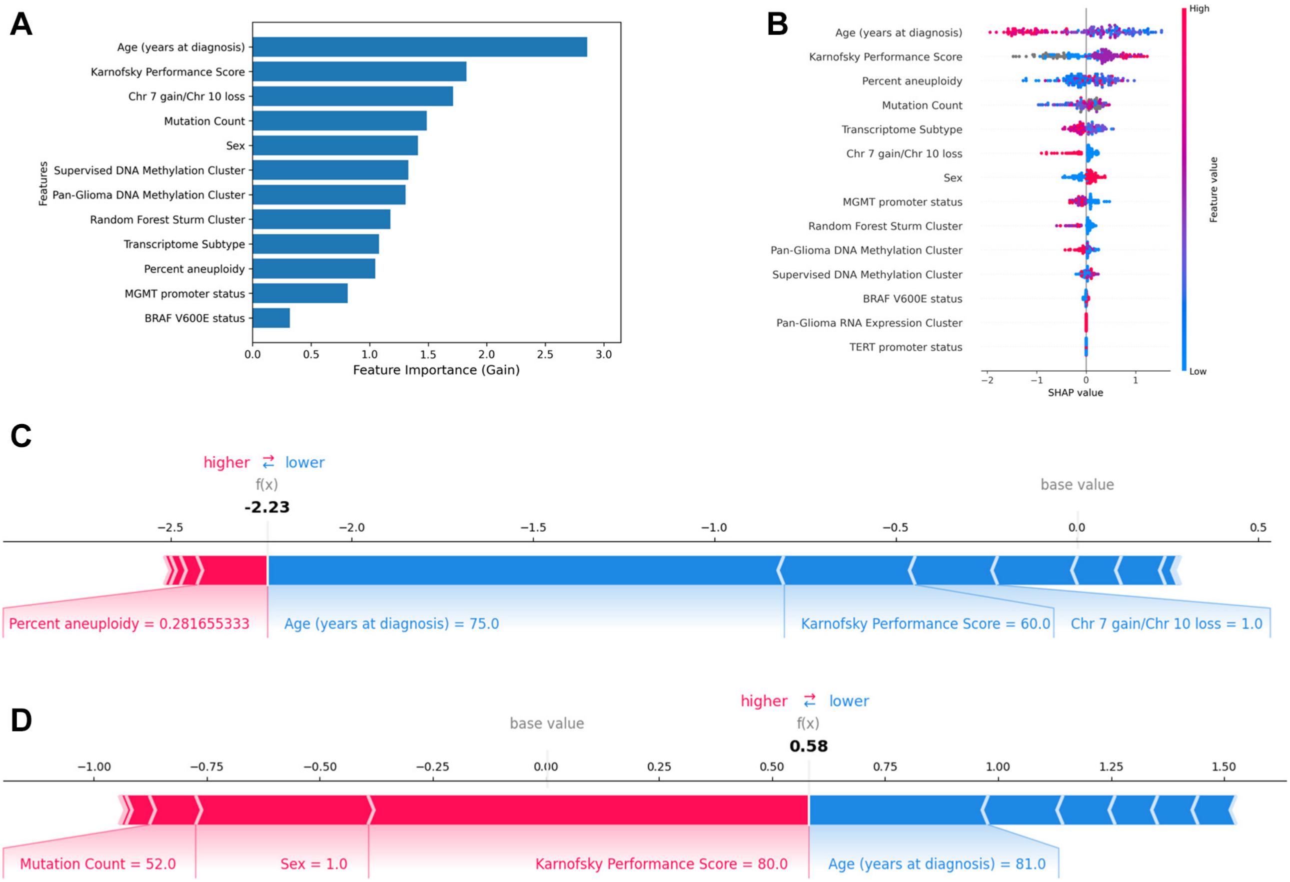

Figure 4.
Analysis of important clinical features as learned by the XGBoost and SHAP. (A) Feature importance with regard to gain that is, overall goodness of a feature to determine the optimal split across all trees. The higher the gain, the more significant the feature while making the prediction. Age has the highest gain followed by Karnofsky Performance Score. (B) SHAP summary plot illustrating all features considered for training XGBoost in decreasing order of impact on the model's output. Gray dots indicate missing values for numeric features. For categoric features except for sex, blue color (indicating "low," missing features are represented as −1 during encoding.) (C) SHAP values for features predictive of a short-survivor patient. (D) SHAP values for features predictive of a long-survivor patient.

finding demonstrates that the proposed approach has the potential to be used for meaningful WSI-level interpretability and visualization in prognostic stratification for clinical research purposes and to identify morphologic features associated with prognosis.

Clinical Analysis: XGBoost and SHAP Analysis

The XGBoost model is widely applicable to machine learning tasks of structured data due to its efficiency, flexibility, and performance. In our study, the model trained on clinical features yielded promising results, underscoring the importance of clinical features in prognostic stratification of patients with GBM. Figure 4A presents the 12 features with nonzero gain, as obtained by the XGBoost model. Feature gain reflects the overall "goodness" of a feature in determining the optimal split across all trees. We illustrated the features in order of decreasing "gain," where a higher gain indicates higher importance of the feature in prognostic stratification.

Age emerged as the most influential continuous feature in prognostic stratification, followed by Karnofsky Performance Score and Chr 7 gain/Chr 10 loss. Interpretability analysis based on SHAP values in Figure 4B illustrates how increasing age negatively affects the prediction of long survivors, whereas a higher Karnofsky Performance Score positively influences the classification of long survivors when studying GBM prognosis. Other significant features include percent aneuploidy, mutation count, and transcriptome subtype. Using SHapley values, we can also observe how different features influence prognostic stratification sample wise. SHAP values for features predictive of short and long survivor are shown in Figure 4C, D, respectively. The length of the bar in these figures shows the magnitude of the feature's effect. As shown in Figure 4C, age (years at diagnosis) contributes the most for classification to a negative class followed by a low Karnofsky Performance Score and Chr 7 gain/Chr 10 loss. Percent aneuploidy has a positive effect on the prediction score; however, the length of the blue bar exceeds the pink bar leading to an overall negative score. For Figure 4D, a high Karnofsky Performance Score, male sex, and mutation count of 50 contribute to classification to long-survivor class. These findings emphasize the significance of these clinical attributes in predicting patient outcomes. It is crucial for correlating imaging data with patient outcomes to better understand the disease and develop superior prognostic models. It is worth noting an intriguing observation in our study—that imaging and clinical data carry complementary information and the models may benefit from further investigation.

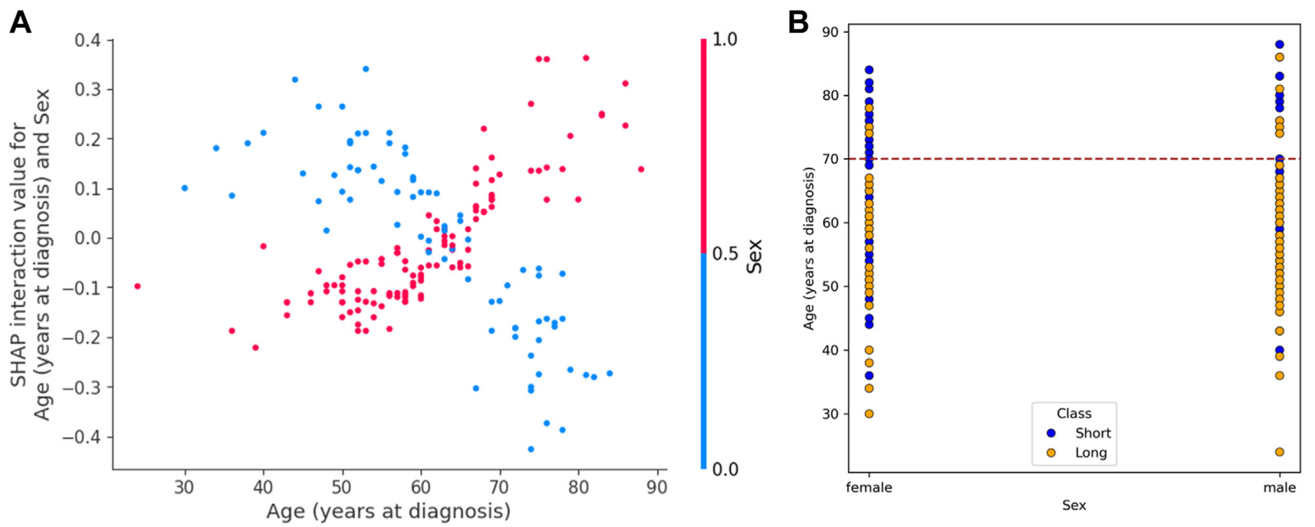

Figure 5.
SHAP interaction effect between age and sex on model outcome. (A) Interaction between age and sex in SHAP. Here, red points represent male (encoded as 1) and blue points indicate female samples (encoded as 0). (B) Distribution of age and sex in the original data.

Subgroup Analysis

The analysis of patient prognosis within specific subgroups holds the potential to unveil critical insights into treatment responses and prognosis variations, thereby paving the way for more personalized treatment strategies in the context of GBM. In this study, we delve into sex-specific subgrouping, conducting evaluations across imaging, clinical, and fusion modalities. From the feature importance analysis in Figure 4B, it is observed that sex, which is a demographic feature, plays a noteworthy role in the model's predictions. Specifically, male sex is associated with a positive effect on the long-survivor class. To delve deeper into the interaction between sex and age on the model's predictions, we conducted a thorough analysis, as shown in Figure 5A. This observation revealed an intriguing pattern: <70 years of age, female sex positively influences the long-survivor class, whereas male sex has a slightly negative impact on long survival. However, this pattern undergoes a substantial shift for patients >70 years of age. We also observed that the male sex has a positive influence on the classification to long-survivor class along with a high Karnofsky Performance Score and low mutation count when age at diagnosis was 81 years (Fig. 4D). Figure 5B provides valuable insights into the distribution of samples concerning age and sex in the data set. Our study cohort included 43 patients aged ≥70 years. Among the 26 female patients in this age group, 3 were classified as long-term survivors, whereas among the 17 male patients, 7 were identified as long-term survivors. Thus, there are comparatively fewer long-survival female patients older than 70 years compared with male patients. The analysis suggests that the SHAP values may disproportionately assign the positive effect of male sex on the model's predictions to the long-survival class. However, for patients younger than 70 years, female patients tend to have better prospects for long survival compared with their male counterparts. This pattern aligns with the findings of our subgroup analysis for sex.

Furthermore, XGBoost models exhibited enhanced performance in the methylated subgroup, both in terms of AUC and accuracy, in contrast to the unmethylated subgroup. Intriguingly, the imaging model demonstrates a higher AUC for the unmethylated subgroup but exhibits lower accuracy when compared with the methylated subgroup. A similar trend manifests in the fusion model results. It is imperative to note, however, that the subgroups categorized under *MGMT* promoter status encompass a limited number of samples, and although trends in the data are noted, generalizable conclusions cannot be drawn. Future investigations on more extensive data sets are essential to substantiate these findings. The variable performance observed across subgroups underscores the significance of considering these subgroup-specific distinctions when interpreting model outcomes.

Despite the recent literature on deep learning–based algorithms in predicting survival outcomes from WSI, these techniques have not yet made a clinical impact by providing the necessary prognostic interpretation. Our approach of prognostic stratification differs from many existing methods in the literature in several ways. First, we followed the latest WHO classification criteria of CNS tumors,[37] which incorporates histologic features, molecular alterations, and genetic alterations in classifying and grading brain tumors to identify the GBM cases. Then, instead of extracting a subset of patches from each WSI or randomly selecting a fixed number of patches for patient-level prognostic decisions,[12,42] we utilize the entire WSI. Although prior studies have emphasized determining OS of patients, we present a more practical and interpretable approach to understand the underlying characteristics that influence OS. We also assessed the associations between demographic and molecular information with patient prognosis in our predictive modeling. Our study cohort includes 15 molecular GBM cases that are particularly challenging to stratify as their histologic features resemble LGG, but the cases are considered as GBM due to their molecular characteristics as per the WHO 2021 guidelines. The stratification performance differed across the 3 models (imaging, clinical, and multimodal). Although each model correctly stratified 8 of 15 cases, their individual correct predictions did not fully overlap indicating complementary strengths. Detailed analysis of 29 incorrectly predicted long survivors revealed that necrotic regions received lower attention compared with other histologic features in these cases, although necrosis is a well-established diagnostic and prognostic feature.[51] An interesting observation is that 15 of these misclassified cases were previously categorized as LGG, having histologic properties of LGG and possible associations with prolonged survival. Furthermore, analysis of the clinical model's predictions reveals a noteworthy outcome that 22 of 29 cases were correctly classified, with only 7 instances of misclassification. This underscores the importance of molecular information in accurately assessing survival outcomes, as it plays a crucial role in distinguishing cases where imaging-based models alone may fall short.

Concluding Remarks, Limitations, and Future Directions

Clinical decision making for patients with glioma is inherently complex and highly personalized, involving a nuanced consideration of factors, such as functional status, tumor aggressiveness, comorbidities, access to care, and treatment options. Our multimodal approach enhances the ability to stratify patients into meaningful prognostic groups, enabling personalized treatment planning. This prognostic information could serve as a valuable tool for the clinical team, supporting decisions on whether to initiate chemotherapy, consider tumor-treating fields, initiate avastin, explore experimental therapies such as clinical trials, and even when to start a treatment versus watch a patient. For example, patients predicted to have a shorter survival could be prioritized for more aggressive or experimental treatments, whereas those with a longer prognosis might benefit from tailored, less-intensive approaches. Moreover, this survival estimate provided by our proposed approach could inform discussions with patients and families, helping to align treatment decisions with individual goals and expectations.

The results of our integrated approach show that data fusion across imaging and clinical models significantly improves the prognostic stratification performance compared with individual data model. This observation underscores the need for data fusion, as it has the potential to mitigate the limitations of individual models and provide a more comprehensive understanding of prognosis in GBM. This approach not only has implications for this specific task but also holds promise for a broader range of applications in the field of health care AI. Qualitative analysis of the obtained results contributes to the interpretability of the algorithmic decisions and provides insights into prognostically relevant characteristics. By identifying these patterns within the current integrated histologic-molecular framework of GBM, clinical neuropathologists can offer additional prognostic insights during microscopic assessments. The results presented in this study also underscore the potential significance of considering sex and *MGMT* subgroups in the stratification of patients with GBM and suggest that sex-specific analyses can augment the precision of patient prognostication. Sex-specific disparities in cancer have remained an underexplored domain, underscoring the need for more studies in this area. In forthcoming research endeavors, the incorporation of sex as a stratification factor in predictive modeling holds promise for improving prognostic accuracy and tailoring treatments to individual patients.

Although this study provides a foundational understanding of prognostic stratification in patients with GBM, it is essential to acknowledge that further investigation is needed to establish the generalizability and robustness for clinical applications. This involves validating the algorithm's performance on diverse data sets with a larger number of cases, including different demographics. The pursuit of an interpretability mechanism to explore and analyze the underlying patterns alongside clinical data and subgroups is an area that warrants further exploration. For instance, there might be hotspots in short-survivor cases that exhibit features of aggressive tumor and determine the outcomes of the patients, whereas the majority displays a less cellular, fibroblastic, spindled appearance that dominates the prediction model and erroneously leads to a prediction of long survival. This underscores the importance of considering the histology of the entire WSI while predicting a patient's prognosis. Also, including multiple WSI from different regions within the tumor could better represent the tumor heterogeneity and may improve the robustness of the analysis by providing deeper insights into the interplay between molecular and histologic characteristics. Although our findings open intriguing avenues for further exploration, it is important to mention that this work does not draw definitive conclusions at this time due to limited sample size. However, the results suggest potential for meaningful insights into GBM prognosis, emphasizing the necessity for further studies to validate and expand upon these initial observations. Conducting a survival analysis to predict time-to-event outcomes would provide a more rigorous framework for understanding survival dynamics rather than relying solely on categoric stratification. Although this is beyond the scope of our current work, we plan to implement survival analysis in future studies to enhance the predictive modeling and generalizability of our findings.

Our study will serve as a foundational stepping stone for future research to deepen the understanding of the intricate interplay between histopathological and clinical features, computational analysis, and clinical outcomes of GBM. Future research can delve into the quantification of the correlation between specific morphologic patterns and OS, shedding light on the intricate relationship between histopathological features and clinical outcomes. We also aim to incorporate additional publicly available data sets, such as IvyGAP[52] and CPTAC,[53] to enhance the diversity and robustness of our training data. Furthermore, we plan to evaluate the impact of domain-specific fine-tuning of ResNet architectures on histopathology images, as demonstrated by Verma et al[54] and compare their performance against standard ImageNet-pretrained counterparts. Additionally, the latest advancements in self-supervised learning models such as vision transformers could be utilized for feature extraction in the current framework. Incorporating self-supervised learning models pretrained on large-scale WSIs could potentially yield contextually rich representations, compared with feature embeddings generated from pretrained ImageNet models, thereby improving the model's discriminative power. As computational approaches in the medical field continue to evolve, integrating additional data such as genomics and radiomics, with histopathological and clinical information could provide a more holistic perspective on GBM. This will pave the way for future studies to explore the combined impact of different data types with a focus on refining and enhancing the interpretability mechanisms of AI models to offer pathologists and clinicians actionable insights for better understanding the rationale behind the underlying disease.

Acknowledgments

B. Baheti, S. Innani, and S. Bakas conducted part of the work reported in this manuscript at their current affiliation, as well as while they were affiliated with the Center for Artificial Intelligence and Data Science for Integrated Diagnostics (AI2D) and the Center for Biomedical Image Computing and Analytics (CBICA) at the University of Pennsylvania, Philadelphia, Pennsylvania.

Author Contributions

B.B. performed conceptualization, data curation, formal analysis, funding acquisition, investigation, methodology, software, validation, visualization, writing – original draft, review, and editing. S.R. performed formal analysis, investigation, methodology, software, validation, visualization, writing – original draft, review, and editing. S.I. performed data curation, formal analysis, investigation, methodology, software, validation, visualization, writing – original draft, review, and editing. G.M. made formal analysis, investigation, methodology, and contributed to writing – review and editing. W.R.B. performed validation and writing – review and editing. S.C.G. contributed to funding acquisition, investigation, methodology, resources, supervision, validation, and writing – review and editing. M.N. performed conceptualization, data curation, investigation, supervision, validation, and writing – review and editing. S.B. contributed to conceptualization, funding acquisition, investigation, methodology, resources, supervision, validation, and writing – review and editing. All authors read and approved the final version of the paper.

Data Availability

Publicly available data sets were analyzed in this study. These data can be found here: https://portal.gdc.cancer.gov/projects/TCGAGBM and https://portal.gdc.cancer.gov/projects/TCGA-LGG.

Funding

Research reported in this publication was partly supported by the Informatics Technology for Cancer Research (ITCR) program of the National Cancer Institute (NCI) of the National Institutes of Health (NIH), under award number U01CA242871, National Institute on Minority Health and Health Disparities of NIH (R01MD018340), NIH grant NCATS/NIH/UL1TR001878, the ITMAT and Penn global engagement research fund of the University of Pennsylvania. The funders had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Declaration of Competing Interest

None reported.

Ethics Approval and Consent to Participate

This study utilized only publicly available, de-identified data sets (TCGA-GBM and TCGA-LGG) that were previously collected with appropriate institutional approvals and patient consent. As no new patient data were collected or analyzed, additional ethical approval and patient consent were not required for this study.

Supplementary Material

The online version contains supplementary material available at https://doi.org/10.1016/j.modpat.2025.100797

References

1. Brennan CW, Verhaak RGW, McKenna A, et al. The somatic genomic landscape of glioblastoma cell. *Cell*. 2013;155:462–477. https://doi.org/10.1016/j.cell.2013.09.034
2. Sottoriva A, Spiteri I, Piccirillo SGM, et al. Intratumor heterogeneity in human glioblastoma reflects cancer evolutionary dynamics. *Proc Natl Acad Sci U S A*. 2013;110:4009–4014. https://doi.org/10.1073/pnas.1219747110
3. Verhaak RGW, Hoadley KA, Purdom E, et al. Integrated genomic analysis identifies clinically relevant subtypes of glioblastoma characterized by abnormalities in PDGFRA, IDH1, EGFR, and NF1. *Cancer Cell*. 2010;17:98–110. https://doi.org/10.1016/j.ccr.2009.12.020
4. Baheti B, Innani S, Nasrallah M, Bakas S. Prognostic stratification of glioblastoma patients by unsupervised clustering of morphology patterns on whole slide images furthering our disease understanding. *Front Neurosci*. 2024;18:1304191. https://doi.org/10.3389/fnins.2024.1304191
5. Redlich J-P, Feuerhake F, Weis J, et al. Applications of artificial intelligence in the analysis of histopathology images of gliomas: a review. *Npj Imaging*. 2024;2:16. https://doi.org/10.1038/s44303-024-00020-8
6. Bakas S, Shukla G, Akbari H, et al. Overall survival prediction in glioblastoma patients using structural magnetic resonance imaging (MRI): advanced radiomic features may compensate for lack of advanced MRI modalities. *J Med Imaging (Bellingham)*. 2020;7:031505. https://doi.org/10.1117/1.JMI.7.3.031505
7. Rathore S, Bakas S, Akbari H, Shukla G, Rozycki M, Davatzikos C. Deriving stable multi-parametric MRI radiomic signatures in the presence of inter-scanner variations: survival prediction of glioblastoma via imaging pattern analysis and machine learning techniques. Paper presented at Houston, Texas, United States, February 27, 2018. Medical Imaging, SPIE Medical Imaging, Computer-Aided Diagnosis; 2018.
8. Macyszyn L, Akbari H, Pisapia JM, et al. Imaging patterns predict patient survival and molecular subtype in glioblastoma via machine learning techniques. *Neuro Oncol*. 2016;18:417–425. https://doi.org/10.1093/neuonc/nov127
9. Beig N, Singh S, Bera K, et al. Sexually dimorphic radiogenomic models identify distinct imaging and biological pathways that are prognostic of overall survival in glioblastoma. *Neuro Oncol*. 2021;23:251–263. https://doi.org/10.1093/neuonc/noaa231
10. Baid U, Rane SU, Talbar S, et al. Overall survival prediction in glioblastoma with radiomic features using machine learning. *Front Comput Neurosci*. 2020;14:61. https://doi.org/10.3389/fncom.2020.00061
11. Lu MY, Williamson DFK, Chen TY, Chen RJ, Barbieri M, Mahmood F. Data-efficient and weakly supervised computational pathology on whole-slide images. *Nat Biomed Eng*. 2021;5:555–570. https://doi.org/10.1038/s41551-020-00682-w
12. Mobadersany P, Yousefi S, Amgad M, et al. Predicting cancer outcomes from histology and genomics using convolutional networks. *Proc Natl Acad Sci*. 2018;115:E2970–E2979. https://doi.org/10.1073/pnas.1717139115
13. Barker J, Hoogi A, Depeursinge A, Rubin DL. Automated classification of brain tumor type in whole-slide digital pathology images using local representative tiles. *Med Image Anal*. 2016;30:60–71. https://doi.org/10.1016/j.media.2015.12.002
14. Cheng J, Mo X, Wang X, Parwani A, Feng Q, Huang K. Identification of topological features in renal tumor microenvironment associated with patient survival. *Bioinformatics*. 2018;34:1024–1030. https://doi.org/10.1093/bioinformatics/btx723
15. Zhu X, Yao J, Huang J. Deep convolutional neural network for survival analysis with pathological images. Paper presented at Shenzhen, China, December 15–18, 2016, 2016 IEEE International Conference on Bioinformatics and Biomedicine (BIBM). IEEE; 2016; New York.
16. Zhu X, Yao J, Zhu F, Huang J. Wsisa: making survival prediction from whole slide histopathological images. Paper presented at New York, July 21 to July 26, 2017. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. IEEE; 2017; New York.
17. Yao J, Zhu X, Jonnagaddala J, Hawkins N, Huang J. Whole slide images based cancer survival prediction using attention guided deep multiple instance learning networks. *Med Image Anal*. 2020;65:101789. https://doi.org/10.1016/j.media.2020.101789
18. Kather JN, Krisam J, Charoentong P, et al. Predicting survival from colorectal cancer histology slides using deep learning: a retrospective multicenter study. *PLoS Med*. 2019;16:e1002730. https://doi.org/10.1371/journal.pmed.1002730
19. Yao J, Wang S, Zhu X, Huang J. Imaging biomarker discovery for lung cancer survival prediction. In: *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer. 2016:649–657.
20. Baheti B, Nasrallah ML, Bakas S, Epco-31. AI-based prognostic stratification of gliosblatoma using H&E-stained whole slide images. *Neuro Oncol*. 2022;24(Supplement_7):vii122-vii122 https://doi.org/10.1093/neuonc/noac209.465
21. Lou C, Habes M, Illenberger N, et al. Leveraging machine learning predictive biomarkers to augment the statistical power of clinical trials with baseline magnetic resonance imaging. *Brain Commun*. 2021;3:fcab264. https://doi.org/10.1093/braincomms/fcab264
22. Chunduru P, Phillips JJ, Molinaro AM. Prognostic risk stratification of gliomas using deep learning in digital pathology images. *Neuro Oncol Adv*. 2022;4:vdac111. https://doi.org/10.1093/noajnl/vdac111
23. Chen RJ, Lu MY, Williamson DFK, et al. Pan-cancer integrative histology-genomic analysis via multimodal deep learning. *Cancer Cell*. 2022;40:865–878.e6. https://doi.org/10.1016/j.ccell.2022.07.004

24. Lipkova J, Chen RJ, Chen B, et al. Artificial intelligence for multimodal data integration in oncology. *Cancer Cell*. 2022;40:1095−1110. https://doi.org/10.1016/j.ccell.2022.09.012
25. Li Y, El Habib Daho M, Conze PH, et al. A review of deep learning-based information fusion techniques for multimodal medical image classification. *Comput Biol Med*. 2024;177:108635. https://doi.org/10.1016/j.compbiomed.2024.108635
26. Vale-Silva LA, Rohr K. Long-term cancer survival prediction using multimodal deep learning. *Sci Rep*. 2021;11:13505. https://doi.org/10.1038/s41598-021-92799-4
27. Steyaert S, Qiu YL, Zheng Y, Mukherjee P, Vogel H, Gevaert O. Multimodal deep learning to predict prognosis in adult and pediatric brain tumors. *Commun Med (Lond)*. 2023;3:44. https://doi.org/10.1038/s43856023-00276-y
28. Cioffi G, Waite KA, Dmukauskas M, et al. Sex differences in glioblastoma response to treatment: impact of MGMT methylation. *Neurooncol Adv*. 2024;6:vdae031. https://doi.org/10.1093/noajnl/vdae031
29. Massey SC, Whitmire P, Doyle TE, et al. Sex differences in health and disease: a review of biological sex differences relevant to cancer with a spotlight on glioma. *Cancer Lett*. 2021;498:178−187. https://doi.org/10.1016/j.canlet.2020.07.030
30. Whitmire P, Rickertsen CR, Hawkins-Daarud A, et al. Sex-specific impact of patterns of imageable tumor growth on survival of primary glioblastoma patients. *BMC Cancer*. 2020;20:447. https://doi.org/10.1186/s12885-020-06816-2
31. Stabellini N, Krebs H, Patil N, Waite K, Barnholtz-Sloan JS. Sex differences in time to treat and outcomes for gliomas. *Front Oncol*. 2021;11:630597. https://doi.org/10.3389/fonc.2021.630597
32. Butler M, Pongor L, Su Y-T, et al. MGMT status as a clinical biomarker in glioblastoma. *Trends Cancer*. 2020;6:380−391. https://doi.org/10.1016/j.trecan.2020.02.010
33. Szylberg M, Sokal P, Śledzińska P, et al. MGMT promoter methylation as a prognostic factor in primary glioblastoma: a single-institution observational study. *Biomedicines*. 2022;10:2030. https://doi.org/10.3390/biomedicines10082030
34. Scarpace L, Mikkelsen L, Cha T, et al. The cancer genome atlas glioblastoma multiforme collection (TCGA-GBM). The Cancer Imaging Archive; 2016. https://doi.org/10.7937/K9/TCIA.2016.rnyfuye9
35. Pedano N, Flanders AE, Scarpace L, et al. The cancer genome atlas low grade glioma collection (TCGA-LGG). The Cancer Imaging Archive; 2016. https://doi.org/10.7937/k9/tcia.2016.l4ltd3tk
36. Clark K, Vendt B, Smith K, et al. The Cancer Imaging Archive (TCIA): maintaining and operating a public information repository. *J Digit Imaging*. 2013;26:1045−1057. https://doi.org/10.1007/s10278-013-9622-7
37. Louis DN, Perry A, Wesseling P, et al. The 2021 WHO classification of tumors of the central nervous system: a summary. *Neuro Oncol*. 2021;23:1231−1251. https://doi.org/10.1093/neuonc/noab106
38. Yang X-L, Zeng Z, Wang C, et al. Predictive model to identify the long time survivor in patients with glioblastoma: a cohort study integrating machine learning algorithms. *J Mol Neurosci*. 2024;74:48. https://doi.org/10.1007/s12031-024-02218-2
39. Biswas A, Salvucci M, Connor K, et al. Comparative analysis of deeply phenotyped GBM cohorts of 'short-term' and 'long-term' survivors. *J Neuro-oncol*. 2023;163:327−338. https://doi.org/10.1007/s11060-023-04341-3
40. Adeberg S, Bostel T, König L, Welzel T, Debus J, Combs SE. A comparison of long-term survivors and short-term survivors with glioblastoma, subventricular zone involvement: a predictive factor for survival? *Radiat Oncol*. 2014;9:1−6. https://doi.org/10.1186/1748-717X-9-95
41. Xu Q-S, Liang Y-Z. Monte Carlo cross validation. *Chemometr Intell Lab Syst*. 2001;56:1−11. https://doi.org/10.1016/S0169-7439(00)00122-2
42. B Ehteshami Bejnordi B, Veta M, Johannes van Diest P, et al. Diagnostic assessment of deep learning algorithms for detection of lymph node metastases in women with breast cancer. *JAMA*. 2017;318:2199−2210. https://doi.org/10.1001/jama.2017.14585
43. Ruifrok AC, Johnston DA. Quantification of histochemical staining by color deconvolution. *Anal Quant Cytol Histol*. 2001;23:291−299.
44. He K, Zhang X, Ren S, Sun J. Deep residual learning for image recognition. Paper presented at New York, June 27−30, 2016. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. IEEE; 2016; New York. https://doi.org/10.48550/arXiv.1512.03385
45. Ilse M, Tomczak J, Welling M. Attention-based deep multiple instance learning. Paper presented at. *Proceedings of the 35th International Conference on Machine Learning80. PMLR*. 2018:2127−2136.
46. Chen T, Guestrin C. Xgboost: a scalable tree boosting system. Paper presented at New York, August 13−17, 2016. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. United States. ACM; 2016; New York.
47. Lundberg SM, Lee S-I. A unified approach to interpreting model predictions. Advances in neural information processing systems. Preprint. Posted online May 22,2017:07874. https://doi.org/10.48550/arXiv.1705.07874. arXiv:1705.
48. Reinke A, Tizabi MD, Baumgartner M, et al. Understanding metric-related pitfalls in image analysis validation. *Nat Methods*. 2024;21:182−194. https://doi.org/10.1038/s41592-023-02150-0
49. Maier-Hein L, Reinke A, Godau P, et al. Metrics reloaded: recommendations for image analysis validation*Nature*. *Nat Methods*. 2024;21:195−212. https://doi.org/10.1038/s41592-023-02151-z
50. Healy J, McInnes L. Uniform manifold approximation and projection. *Nat Rev Methods Primers*. 2024;4:82. https://doi.org/10.1038/s43586-024-00363-x
51. Barker FG, Davis RL, Chang SM, Prados MD. Necrosis as a prognostic factor in glioblastoma multiforme. *Cancer*. 1996;77:1161−1166. https://doi.org/10.1002/(sici)1097-0142(19960315)77:6<1161::aid-cncr24>3.0.co;2-z
52. Puchalski RB, Shah N, Miller J, et al. An anatomic transcriptional atlas of human glioblastoma. *Science*. 2018;360:660−663. https://doi.org/10.1126/science.aaf2666
53. Liu J, Cao S, Imbach KJ, et al. Multi-scale signaling and tumor evolution in high-grade gliomas. *Cancer Cell*. 2024;42:1217−1238.e19. https://doi.org/10.1016/j.ccell.2024.06.004
54. Verma R, Alban TJ, Parthasarathy P, et al. Sexually dimorphic computational histopathological signatures prognostic of overall survival in high-grade gliomas via deep learning. *Sci Adv*. 2024;10:eadi0302. https://doi.org/10.1126/sciadv.adi0302